

QUESTION 1 (9 Marks)	Marks
(a) (i) How many ways could the letters of the word SOCIETY be arranged if each arrangement begins with C and ends with E.	1
(ii) If an arrangement is selected at random, find the probability that it contains the word SOY.	1
(b) The displacement function of a particle moving x metres along a straight line after t seconds is given by $x = \sqrt{2} \cos 5t - \sin 5t$. Show that its acceleration function is of the form $\ddot{x} = -n^2x$ and find the value of n .	2
(c) A plane travelling at a constant height of 1500 metres at a speed of 600 km/hr releases a bomb. What is the horizontal distance the bomb has travelled when it hits the ground. (Take $g=10 \text{ m/s}^2$).	3
(d) (i) How many different ways could four cards be selected from a regular pack of 52 playing cards.	1
(ii) How many of these selections will contain exactly two Aces.	1

QUESTION 2 (9 Marks)

(a) A sky-diver opens his parachute when falling at 30 m/s. Thereafter, his acceleration is given by $\frac{dv}{dt} = k(6 - v)$, where k is a constant.	
(i) Show that this condition is satisfied when $v = 6 + Ae^{-kt}$, and find the value of A .	2
(ii) One second after opening his parachute, his velocity has fallen to 10.7 m/s. Find k to two decimal places.	2
(iii) Find, to one decimal place, his velocity two seconds after his parachute has opened.	2
(iv) If, with the same acceleration, the sky-diver opens his parachute when falling at 6 m/s, briefly describe his subsequent motion.	1
(b) Persons A, B, C, D, E, F and G are to be seated at a round table. How many arrangements are possible if A refuses to sit next to B or C .	2

QUESTION 3 (9 Marks)

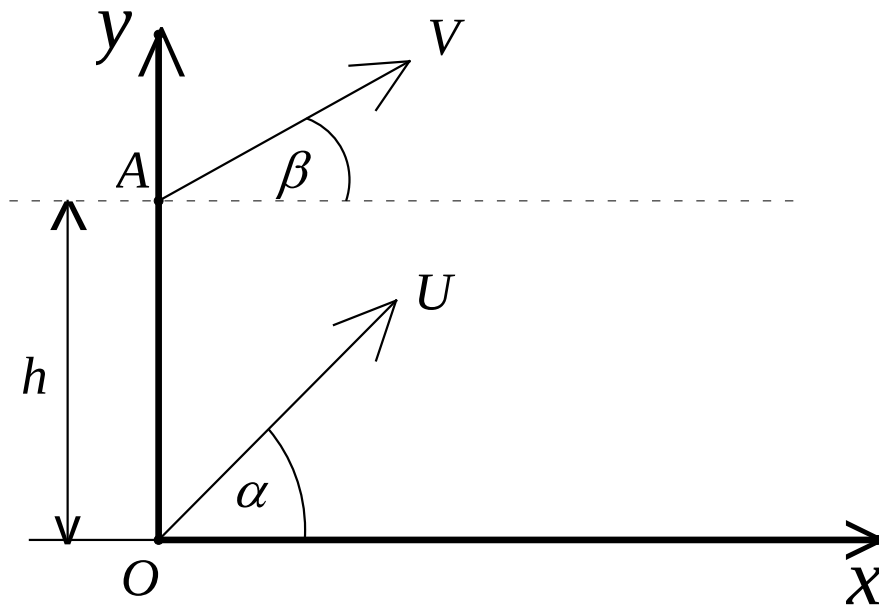
- (a) A particle moves in Simple Harmonic Motion. When it is 2 metres and 3 metres respectively from its centre of motion, its velocity is respectively 6 m/s and 4 m/s. Find the period of its motion and its amplitude. 3
- (b) A function $N(t)$ is given by $N(t) = Ae^{\frac{t}{3}} + Be^{-\frac{2t}{3}}$, where A and B are constants.
- (i) If $N(0) = 30$ and $N'(0) = -14$, find A and B . 2
- (ii) Find, to 2 decimal places, the value of t for which $N(t)$ is a minimum, and find this minimum value. 3
- (iii) Briefly describe the behaviour of $N(t)$ as t increases. 1

QUESTION 4 (9 Marks)

- (a) How many arrangements of the letters of the word CONTAINER are possible if:
- (i) there are no restrictions. 1
- (ii) the vowels are together. 1
- (b) In a herd of 500 cows, the number N infected with a disease at time t years is given by $N = \frac{500}{1 + Ae^{-500t}}$.
- (i) Briefly explain why all the cows will eventually be infected. 1
- (ii) Initially, only one cow was infected. After how many days will 200 cows be infected. 3
- (iii) Show that $\frac{dN}{dt} = N(500 - N)$. 3

QUESTION 5 (9 Marks)

- (a) The equation of motion of a particle moving in Simple Harmonic Motion is given by $x = a \cos (nt + \alpha)$, where x metres is its displacement from origin O after t minutes. It is initially 6 metres right of O and moving towards it. The period of its motion is 8 minutes and its maximum speed is 3π m/min. Find:
- (i) the values of n , a and α . 3
- (ii) the first time when it passes through the origin. 1
- (b) In the diagram below, a particle is projected from the origin O with a speed of U m/s at an angle of elevation α . At the same instant, another particle is projected from A , h metres above O with a speed of V m/s at an angle of elevation β ($\beta < \alpha$). The particles move in the same plane of motion and collide T seconds after projection. 5



The horizontal and vertical components of displacement t seconds after the particle is projected from O are given by $x_o = Ut \cos \alpha$ and $y_o = Ut \sin \alpha - \frac{1}{2}gt^2$ respectively, and the horizontal and vertical components of displacement t seconds after the particle is projected from A are given by $x_A = Vt \cos \beta$ and $y_A = h + Vt \sin \beta - \frac{1}{2}gt^2$ respectively.

Show that $T = \frac{h \cos \beta}{U \sin(\alpha - \beta)}$.

QUESTION 6 (9 Marks)

- (a) The displacement function of a particle moving x metres along a straight line after t seconds is given by $x = 3\cos^2 4t$. Show that its motion is Simple Harmonic and find its centre of motion. 3
- (b) The acceleration of a particle moving along a straight line is given by $\ddot{x} = 3x(x - 2)$, where x metres is its displacement from the origin O after t seconds. Initially it is at O and its velocity is 2 m/s .
- (i) Show that $v = 2(x^3 - 3x^2 + 2)$, where v is its velocity. 2
- (ii) Find its velocity and acceleration at $x = 1$. 2
- (iii) Briefly describe its motion after it moves from $x = 1$. 2

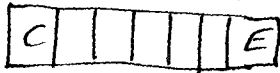
QUESTION 7 (9 Marks)

- (a) A velocity function is given by $\frac{dx}{dt} = (4 - 3x)^2$. Find $\frac{d^2x}{dt^2}$. 2
- (b) A team of FIVE is to be selected from a group of FOUR boys and FOUR girls.
- (i) How many teams are possible if there is to be a majority of girls. 1
- (ii) What is the probability of a particular girl being included in the team and a particular boy not included, still assuming a majority of girls in the team. 2
- (c) On a certain day, the depth of water in a harbour at high tide is 11 metres. 4
At low tide $6\frac{1}{4}$ hours later, the depth of water is 7 metres. If high tide is due at 2.50 AM, what is the earliest time after midday that a ship requiring a depth of at least 10 metres of water can enter the harbour.

Solutions to Year 12 Term 2 Assessment 2007

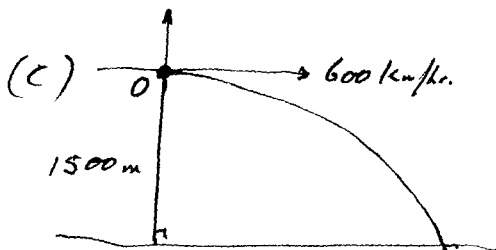
Question 1

(a)(i) SOCIETY



$$5! = 120$$

$$(ii) p = \frac{5!}{7!} = \frac{1}{42}$$



$$V = 600 \text{ km/hr} \\ = 600 \times \frac{1000}{3600} \text{ m/s} \\ = \frac{500}{3} \text{ m/s}$$

$$(b) x = \sqrt{2} \cos 5t - \sin 5t \quad (1)$$

$$\dot{x} = -5\sqrt{2} \sin 5t - 5 \cos 5t$$

$$\ddot{x} = -5^2 \sqrt{2} \cos 5t + 5^2 \sin 5t \\ = -5^2 (\sqrt{2} \cos 5t - \sin 5t)$$

$$\therefore \ddot{x} = -5^2 x \text{ from (1)}$$

This is of the form $\ddot{x} = -n^2 x$ where $n = 5$.

$$\ddot{x} = 0$$

$$\dot{x} = V \cos \alpha$$

$$\therefore \dot{x} = V$$

$$\therefore x = Vt$$

Eliminating t :

$$y = -\frac{g x^2}{2 V^2}$$

Taking 0 as origin, $g = 10$, $x = ?$, $y = -1500$ and $V = \frac{500}{3}$

$$\therefore +1500 = +\frac{10 x^2}{2} \div \left(\frac{500}{3}\right)^2$$

$$\therefore x^2 = \frac{500 \times 500 \times 100}{3} \\ = \frac{25 \times 10^6}{3}$$

$$\therefore x = \frac{5000\sqrt{3}}{3} \text{ metres or } \frac{5\sqrt{3}}{3} \text{ km.}$$

$$(iii) \text{ When } t = 2 \\ v = 6 + 24 e^{-2 \times 1.63}$$

$$\therefore v = 6.9 \text{ m/s (1 dec. pt)}$$

(iv) Acceleration is $\frac{dv}{dt} = K(6-v)$. When $v = 6$, $\frac{dv}{dt} = 0$ $\therefore$ he goes at a constant rate of 6 m/s.

Question 2

$$(a)(i) v = 6 + A e^{-Kt}$$

$$\frac{dv}{dt} = -K \times A e^{-Kt}$$

$$= -K(v-6)$$

$$\therefore \frac{dv}{dt} = K(6-v)$$

$$\text{When } t=0, v=30$$

$$\therefore 30 = 6 + A$$

$$\therefore A = 24$$

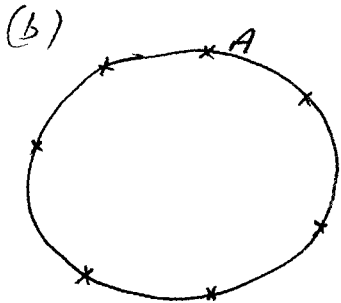
$$(ii) \text{ When } t=1, v=10.7$$

$$\therefore 10.7 = 6 + 24 e^{-K}$$

$$\therefore \frac{4.7}{24} = e^{-K}$$

$$\therefore K = \ln \frac{24}{4.7}$$

$$\therefore K = 1.63 \text{ (2 dec. pt.)}$$



Seat A. Now B has 4 choices and C has 3. The remainder can be seated in $4!$ ways. $\therefore 4! \times 4 \times 3 = 288$ arrangements.

Question 3

(a) Using $v^2 = n^2(a^2 - x^2)$
 $\therefore 6^2 = n^2(a^2 - 2^2)$ and $4^2 = n^2(a^2 - 3^2)$
 $\therefore 36 = n^2(a^2 - 4)$ $16 = n^2(a^2 - 9)$

$$\therefore \frac{36}{a^2 - 4} = \frac{16}{a^2 - 9}$$

$$\therefore 36a^2 - 324 = 16a^2 - 64$$

$$\therefore 20a^2 = 260$$

$$\therefore a = \sqrt{13} \quad (a > 0)$$

(b) $N(t) = A e^{\frac{t}{3}} + B e^{-\frac{2t}{3}}$

(i) $\therefore N'(t) = \frac{A}{3} e^{\frac{t}{3}} - \frac{2B}{3} e^{-\frac{2t}{3}}$

When $t = 0$

$$A + B = 30 \quad \text{--- (1)}$$

$$\frac{A}{3} - \frac{2B}{3} = -14 \quad \text{--- (2)}$$

$$(2): A - 2B = -42$$

$$\therefore A - 2(30 - A) = -42 \quad \text{from (1)}$$

$$\therefore A = 6$$

$$B = 24$$

(iii) As $t \rightarrow \infty$, $e^{\frac{t}{3}} \rightarrow \infty$ and $e^{-\frac{2t}{3}} \rightarrow 0$. $\therefore N(t) \rightarrow \infty$.

Question 4

(a)(i) CONTAINER

$$\frac{9!}{2!} = 181440$$

(ii) $(AEIO) \times \times \times \times \times$

$$\frac{6!4!}{2!} = 8,640$$

(b)(i) $N = \frac{500}{1 + A e^{-500t}}$

As t increases, $e^{-500t} \rightarrow 0 \therefore N \rightarrow 500$

(iii) $N = 500(1 + A e^{-500t})^{-1}$

$$\therefore \frac{dN}{dt} = -500(1 + A e^{-500t})^{-2} \times -500A e^{-500t}$$

$$= \frac{500^2 A e^{-500t}}{(1 + A e^{-500t})^2} \times A e^{-500t}$$

$$= N^2 \times A e^{-500t}$$

$$= N^2 \left(\frac{500}{N} - 1 \right) \text{ since } 1 + A e^{-500t} = \frac{500}{N}$$

$$\therefore \frac{dN}{dt} = N(500 - N)$$

$$\text{Let } 4^2 = n^2(a^2 - 9)$$

$$\therefore 16 = n^2(13 - 9)$$

$$\therefore n = 2 \quad (n > 0)$$

$$\text{Now } T = \frac{2\pi}{n} = \frac{2\pi}{2}$$

$$\therefore T = \pi \text{ sec.}$$

$\therefore$ the period of its motion is π sec. and its amplitude is $\sqrt{13}$ metres.

(ii) When $N'(t) = 0$, $\frac{A}{3} e^{\frac{t}{3}} = \frac{2B}{3} e^{-\frac{2t}{3}}$
 $\therefore 2 e^{\frac{t}{3}} = 16 e^{-\frac{2t}{3}}$

$$\therefore e = 8$$

$$\therefore t = 2.08 \quad \frac{2.08}{3} \quad -\frac{2}{3} \times 2.08$$

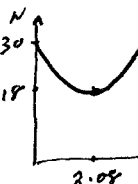
$$\text{When } t = 2.08, N = 6e^{\frac{2.08}{3}} + 24e^{-\frac{2 \times 2.08}{3}}$$

$$\therefore N = 18$$

Test.

$$N''(t) = \frac{A}{9} e^{\frac{t}{3}} + \frac{4B}{9} e^{-\frac{2t}{3}}$$

$$\text{At } t = 2.08, N''(t) = \frac{2}{3} e^{\frac{2.08}{3}} + \frac{32}{3} e^{-\frac{2 \times 2.08}{3}} = 3.99 > 0 \therefore N$$



Question 5

(a)(i) $x = a \cos(nt + \alpha)$

When $t=0$, $x=6$

$\therefore 6 = a \cos \alpha$

Since $T = \frac{2\pi}{n} = 8$

$\therefore n = \frac{\pi}{4} \quad \text{--- (1)}$

Now $v^2 = n^2 (a^2 - x^2)$

Since v is max. when $x=0$

$\therefore v^2 = n^2 a^2 \therefore v = \pm na$

Initially $x = +6 \therefore v = +na$

$\therefore 3\pi = \frac{\pi}{4} a \therefore a = 12 \quad \text{--- (2)}$

Now $6 = 12 \cos \alpha \therefore \cos \alpha = \frac{1}{2}$

$\therefore \alpha = \frac{\pi}{3} \quad \text{--- (3)}$

Since $x = a \cos(nt + \alpha)$

$\therefore x = 12 \cos\left(\frac{\pi t}{4} + \frac{\pi}{3}\right)$

When $x=0$

$0 = 12 \cos\left(\frac{\pi t}{4} + \frac{\pi}{3}\right)$

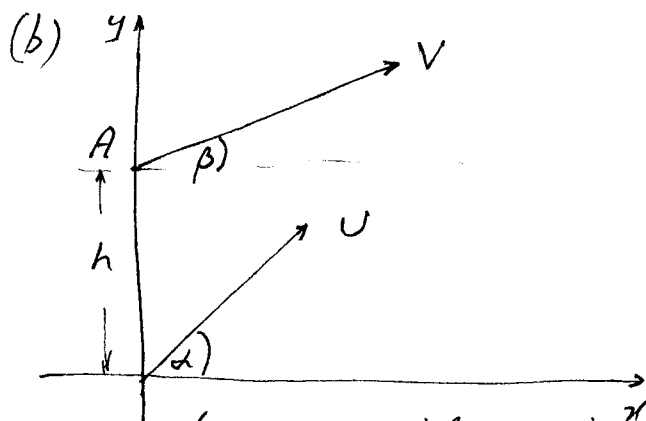
$\therefore \cos \frac{\pi}{2} = 12 \cos\left(\frac{\pi t}{4} + \frac{\pi}{3}\right)$ by initial conditions

$\therefore \frac{\pi t}{4} + \frac{\pi}{3} = \frac{\pi}{2}$

$\therefore \frac{\pi t}{4} = \frac{\pi}{6}$

$\therefore t = \frac{2}{3}$

$\therefore$ it passes through the origin first time when $t = \frac{2}{3}$ minutes.



The particles will collide when $x_0 = x_A$

$\therefore Ut \cos \alpha = Vt \cos \beta$

$\therefore U \cos \alpha = V \cos \beta \quad \text{--- (1)}$

When $t = T$, $y_0 = y_A$

$\therefore UT \sin \alpha - \frac{1}{2}gt^2 = h + Vt \sin \beta - \frac{1}{2}gt^2$

$\therefore T(U \sin \alpha - V \sin \beta) = h$

$\therefore T = \frac{h}{U \sin \alpha - V \sin \beta}$

$= \frac{h}{U \sin \alpha - V \cos \alpha \sin \beta}$

$\text{From (1) } V \cos \alpha = U \sin \alpha$

$= \frac{h}{U \sin \alpha - U \cos \alpha \sin \beta}$

$\therefore T = \frac{h \cos \beta}{U \sin(\alpha - \beta)}$

Question 6

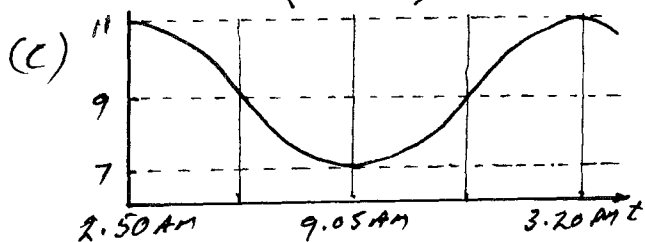
(a) $x = 3 \cos^2 4t$ — (1)
 Now $\cos 8t = 2 \cos^2 4t - 1$ — (2)
 $\therefore \cos^2 4t = \frac{1}{2}(1 + \cos 8t)$
 $\therefore x = \frac{3}{2}(1 + \cos 8t)$
 $\therefore \dot{x} = -12 \sin 8t$
 $\therefore \ddot{x} = -96 \cos 8t$
 $= -96\left(\frac{2x}{3} - 1\right)$ from (1) and (2)
 $\therefore \ddot{x} = -8^2\left(x - \frac{3}{2}\right)$

Since, of the form $\ddot{x} = -n^2(x - x_0)$ $\therefore$ SHM
 and centre of motion is
 at $x = \frac{3}{2}$, i.e., $\frac{3}{2}$ metres
 right of origin.

Question 7

(a) $v = \frac{dx}{dt} = (4 - 3x)^2$
 Now $\ddot{x} = \frac{d^2x}{dt^2} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$

$\therefore \ddot{x} = \frac{d}{dx}\left(\frac{1}{2}(4 - 3x)^4\right)$
 $= \frac{4}{2}(4 - 3x)^3 \times -3$
 $\therefore \ddot{x} = -6(4 - 3x)^3$



Let $x = a + b \cos(nt + \alpha)$
 Since $a = 9$ and $b = 2$
 $\therefore x = 9 + 2 \cos(nt + \alpha)$
 When $t = 0$, $x = 11$
 $\therefore 11 = 9 + 2 \cos \alpha$
 $\cos \alpha = 1 \therefore \alpha = 0$
 also $T = \frac{2\pi}{n} = 12\frac{1}{2} = \frac{25}{2}$
 $\therefore n = \frac{4\pi}{25}$

(b) $\ddot{x} = v \frac{dv}{dx} = 3x(x - 2)$

(i) $\therefore v dv = (3x^2 - 6x) dx$
 $\therefore \int v dv = \int (3x^2 - 6x) dx$
 $\therefore \frac{1}{2}v^2 = x^3 - 3x^2 + C$
 when $x = 0$, $v = 2 \therefore C = 2$
 $\therefore v^2 = 2(x^3 - 3x^2 + 2)$

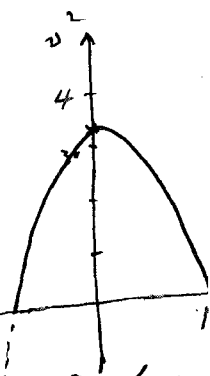
(ii) At $x = 1$, $v^2 = 2(1 - 3 + 2)$
 $\therefore v = 0$

and $\ddot{x} = 3(1 - 2)$
 $= -3 \text{ m/s}^2$

(iii)

x	-2	-1	0	1	2	3
v^2	-36	-4	4	0	-2	

At $x = 1$, force is -ve,
 $\therefore$ particle moves towards 0
 with increasing speed.
 At $x = 0$, particle slows
 down and stops between $x = 0$ and $x = 1$.
 It oscillates between this point
 and $x = 1$, but not in SHM.



(b)(i) We require $2B + 3G$, $1B + 4G$
 $\therefore {}^4C_2 \times {}^4C_3 + {}^4C_1 \times {}^4C_4$
 $= 28$

(ii) $P = \frac{{}^3C_1 \times {}^3C_3 + {}^3C_2 \times {}^3C_2}{28}$
 $= \frac{12}{18} \text{ or } \frac{3}{7}$

$\therefore x = 9 + 2 \cos \frac{4\pi t}{25}$

When $x = 10$

$\therefore 10 = 9 + 2 \cos \frac{4\pi t}{25}$

$\therefore \cos \frac{4\pi t}{25} = \frac{1}{2} = \cos \frac{\pi}{3}$

$\therefore \frac{4\pi t}{25} = 2n\pi \pm \frac{\pi}{3}$

Let $n = 1$ and consider $-\frac{\pi}{3}$

$\frac{4\pi t}{25} = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$

$\therefore t = 10.42 \text{ hrs or } 10 \text{ hrs } 25 \text{ minutes.}$

Now 2.50 AM + 10 hrs 25 min is 1.15 PM
 $\therefore$ earliest time is 1.15 PM